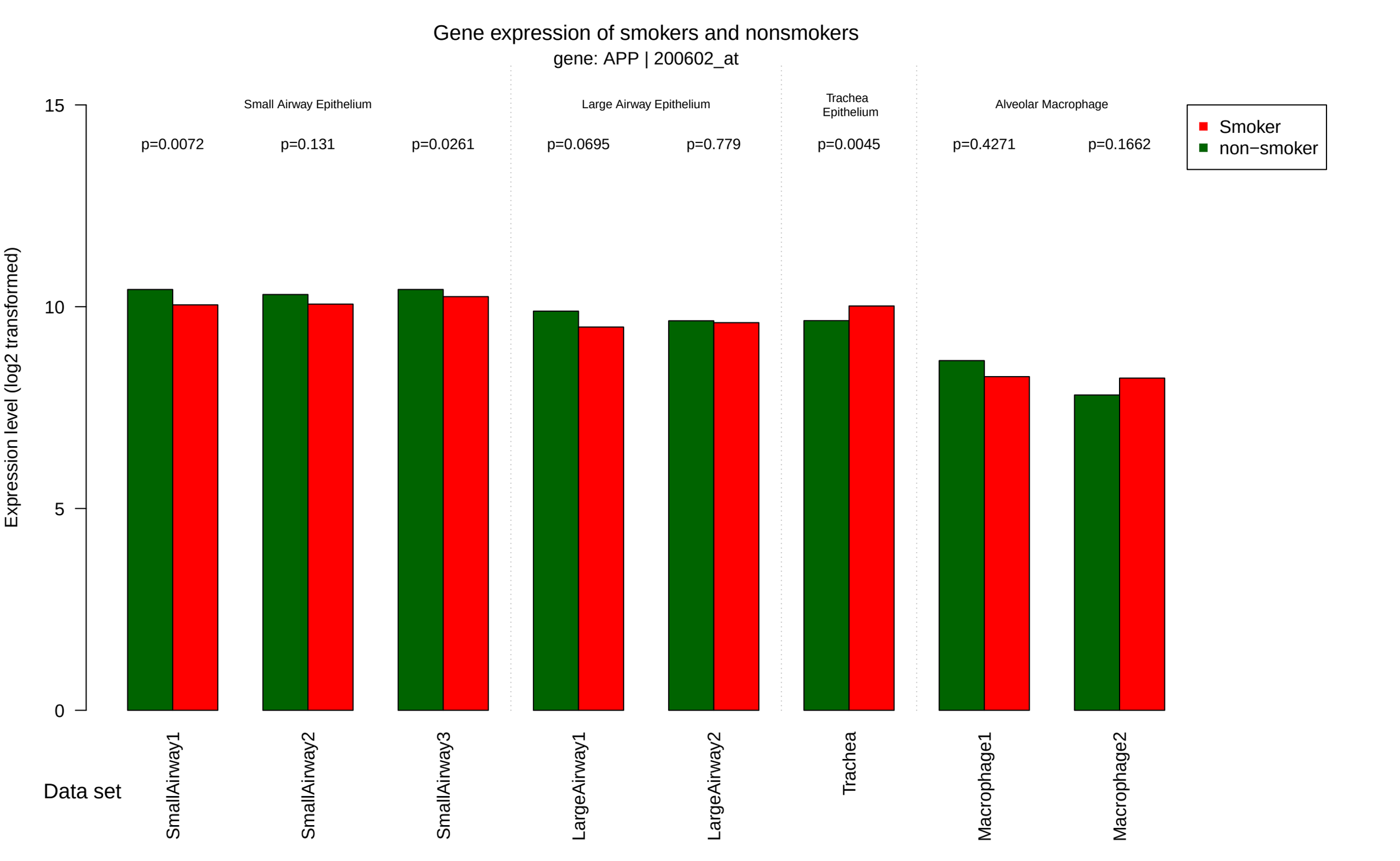
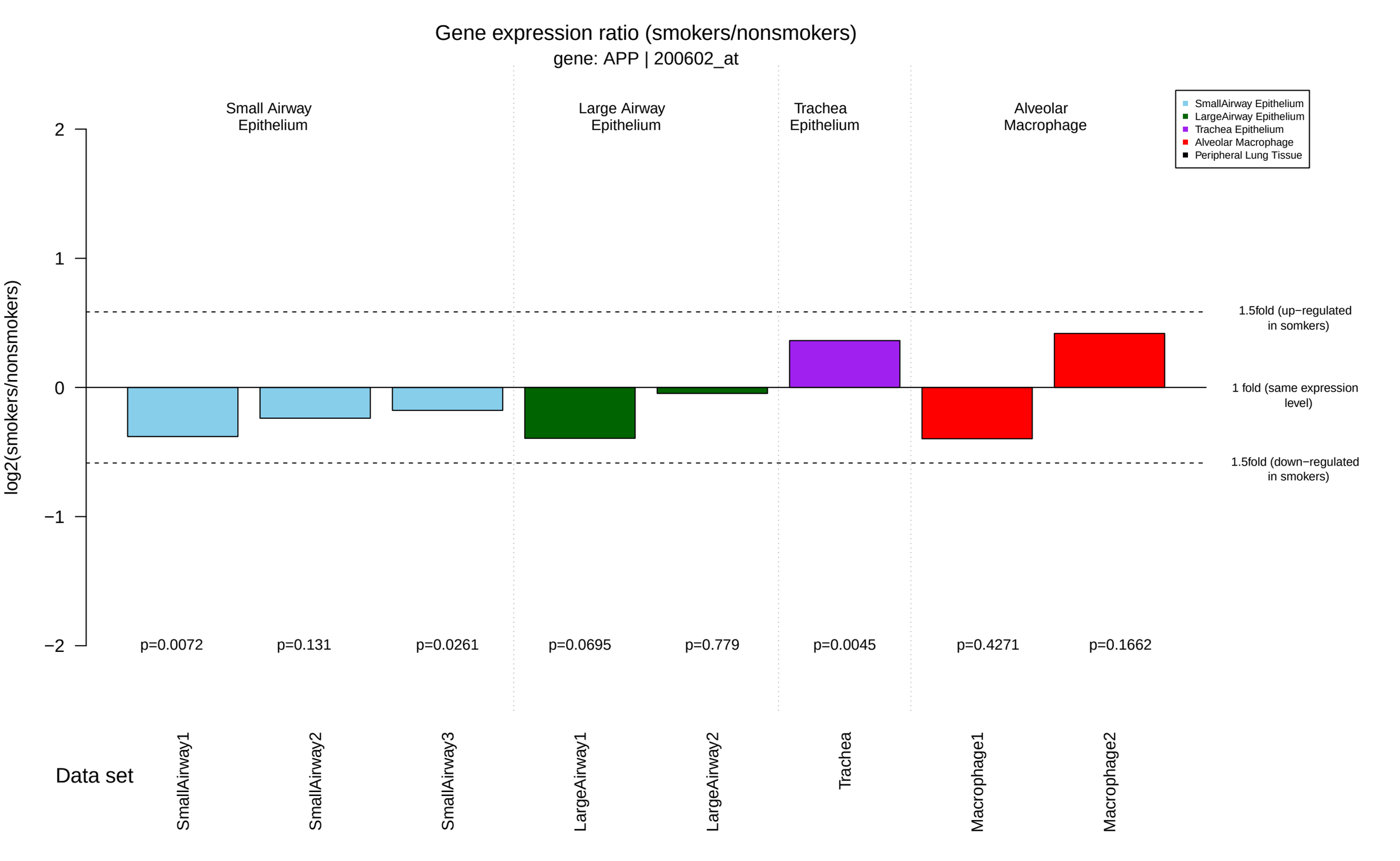
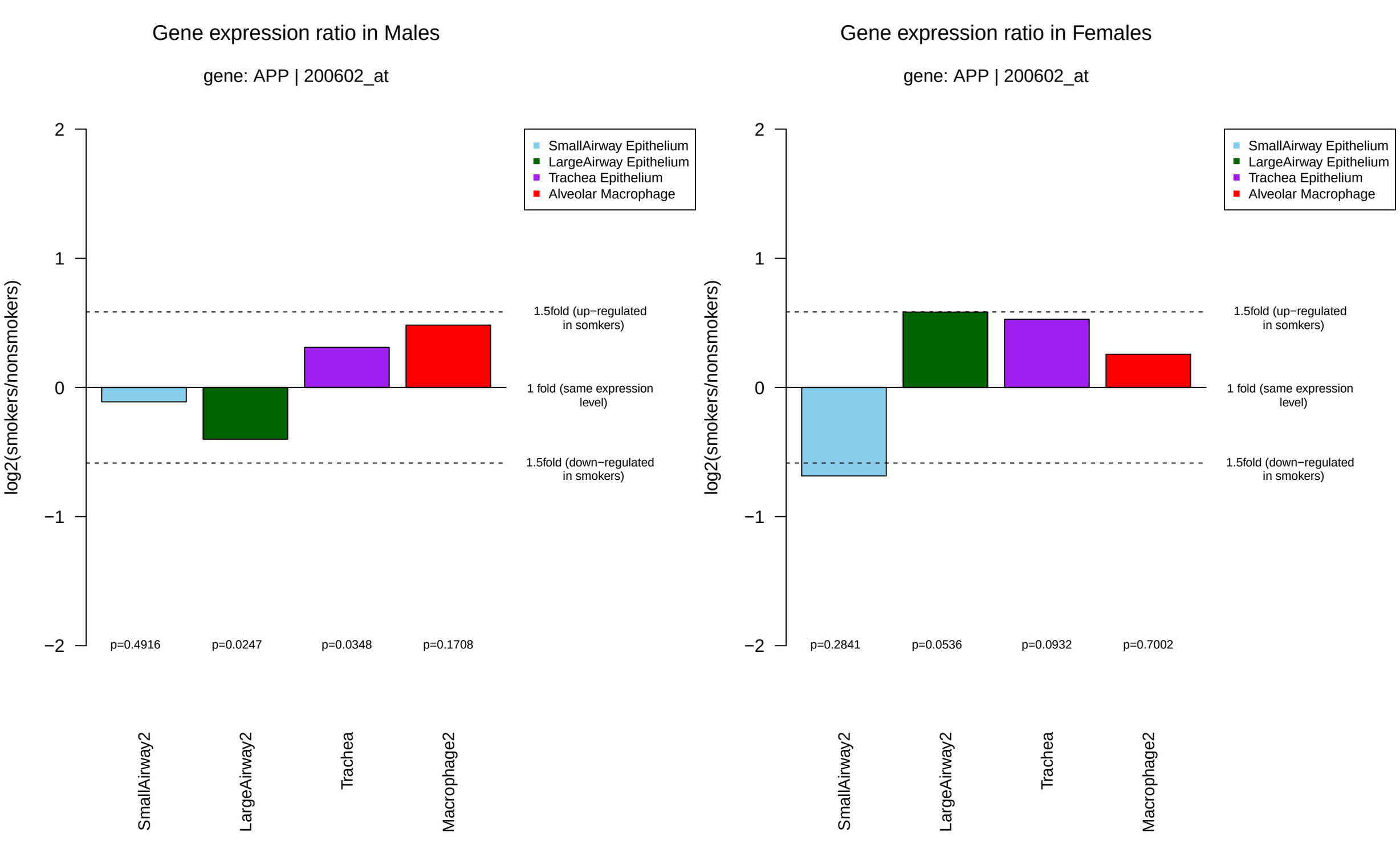
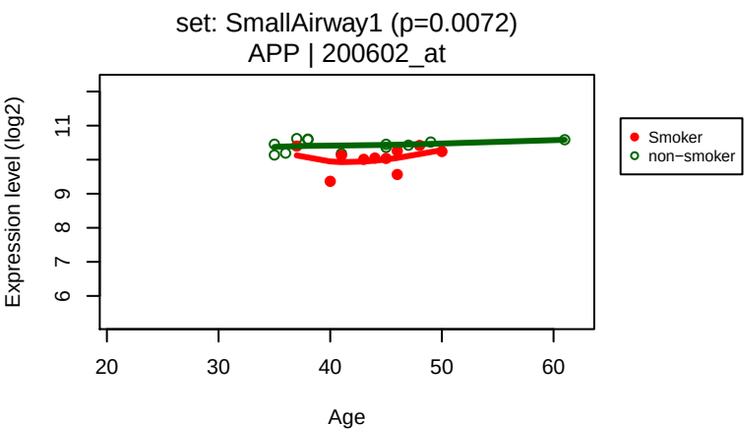
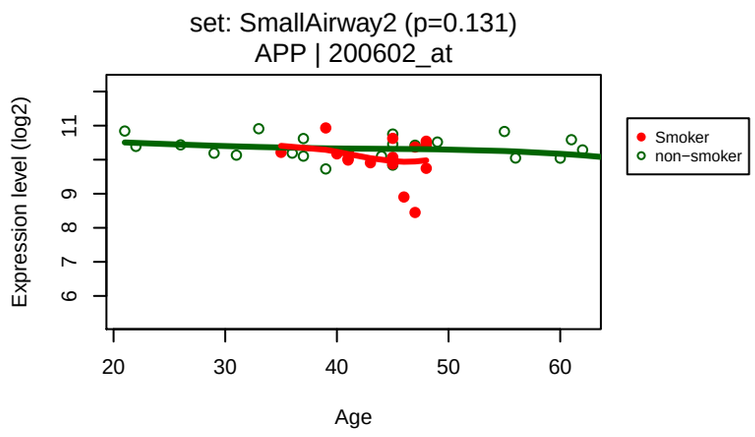
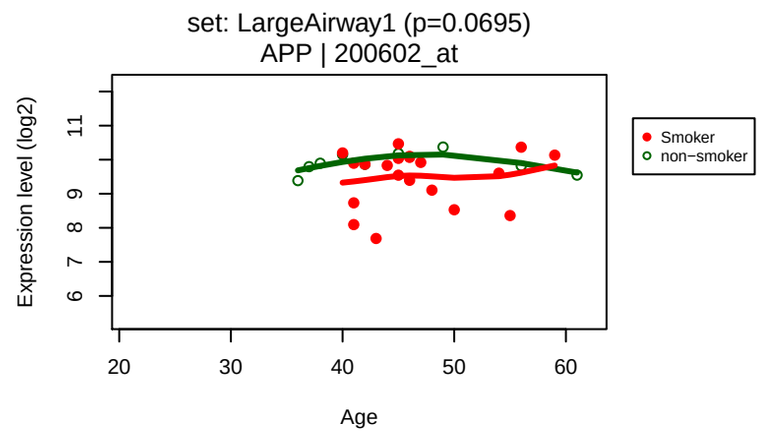

Gene expression of smokers and nonsmokers
gene: APP | 200602_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0072
p=0.131
p=0.0261
p=0.0695
p=0.779
p=0.0045
p=0.4271
p=0.1662
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

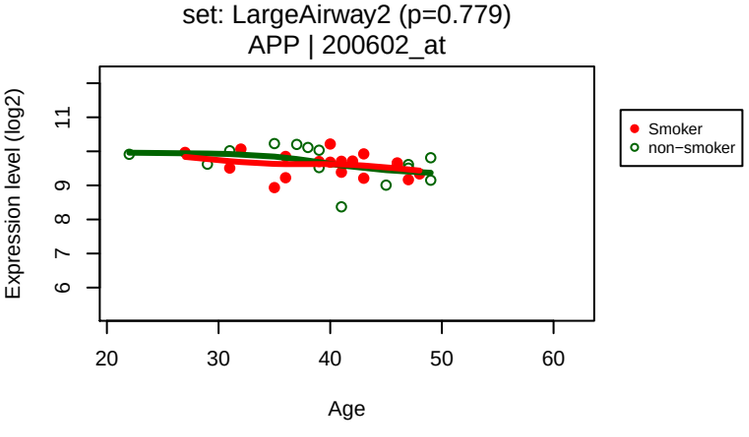

Gene expression ratio (smokers/nonsmokers)
gene: APP | 200602_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0072
p=0.131
p=0.0261
p=0.0695
p=0.779
p=0.0045
p=0.4271
p=0.1662
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

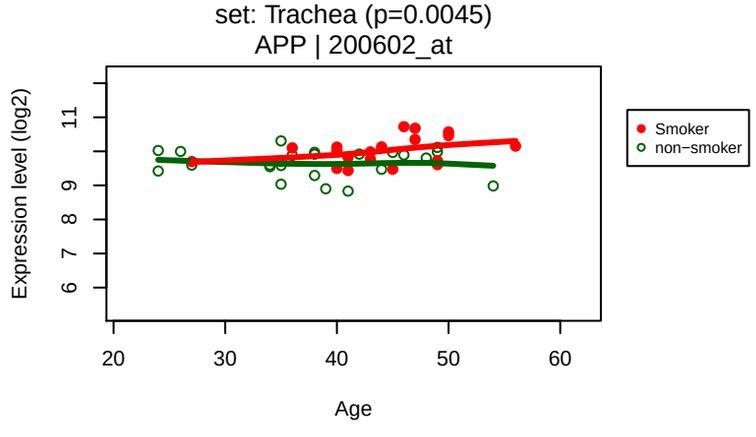

Gene expression ratio in Males
gene: APP | 200602_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4916
p=0.0247
p=0.0348
p=0.1708
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: APP | 200602_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2841
p=0.0536
p=0.0932
p=0.7002
SmallAirway2
LargeAirway2
Trachea
Macrophage2

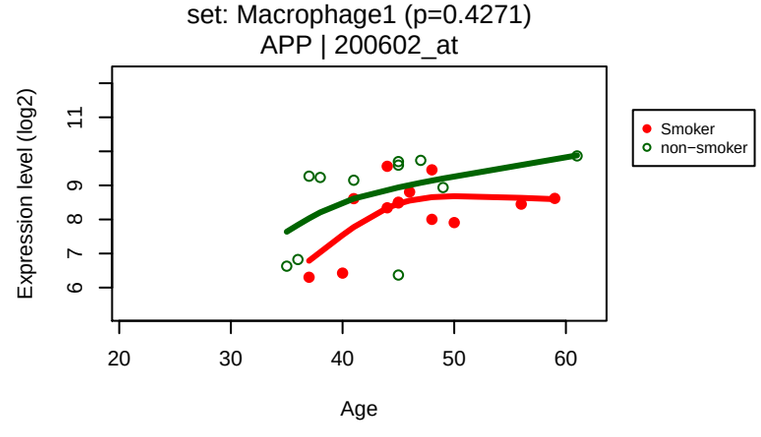

set: SmallAirway1 (p=0.0072)
APP | 200602_at
Expression level (log2)
Age
Smoker
non−smoker

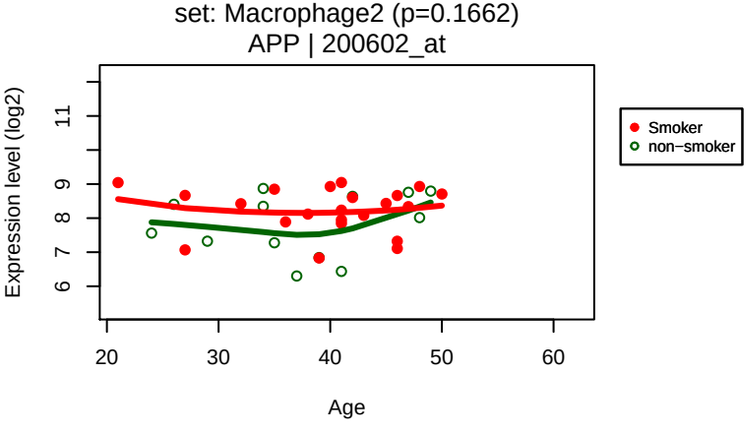

set: SmallAirway2 (p=0.131)
APP | 200602_at
Expression level (log2)
Age
Smoker
non−smoker

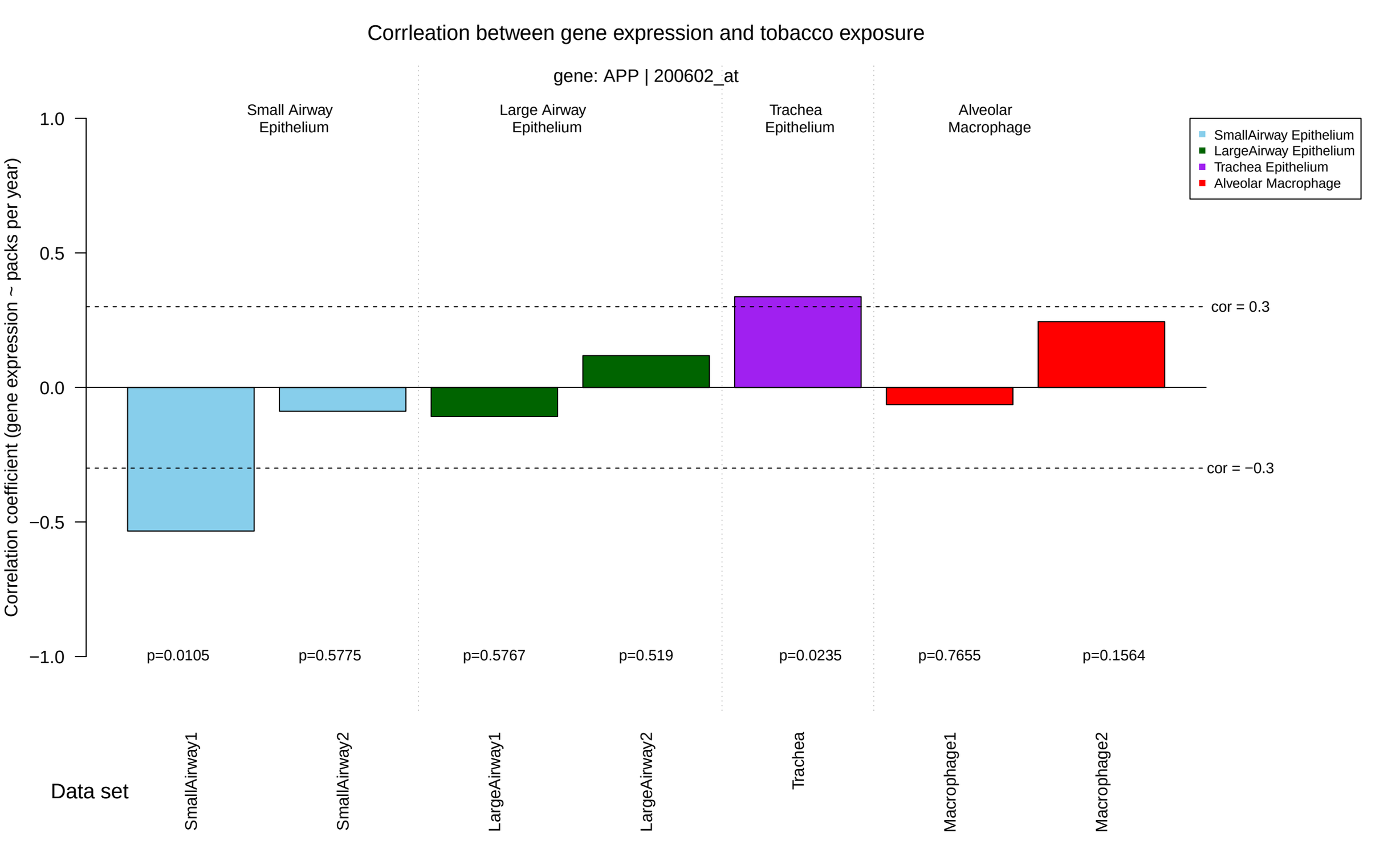

set: LargeAirway1 (p=0.0695)
APP | 200602_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.779)
APP | 200602_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0045)
APP | 200602_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4271)
APP | 200602_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1662)
APP | 200602_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: APP | 200602_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0105
p=0.5775
p=0.5767
p=0.519
p=0.0235
p=0.7655
p=0.1564
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2